

# Shelby Ziccardi

PH.D. CANDIDATE, UNIVERSITY OF MINNESOTA - TWIN CITIES

RESEARCH INTERESTS: THE CAPTURE AND SIMULATION OF NATURAL MOVEMENT

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## Skills

**Software** Unity, Maya, Blender, Adobe Creative Suite  
**Libraries** OpenGL, OpenCV, OpenPose, Eigen, PyTorch, Matplotlib, Numpy, PyShiny, Streamlit  
**Languages** C++, C, C#, Python, Java, GLSL, LaTeX

## Education

### University of Minnesota - Twin Cities

Minneapolis, MN

PH.D. IN COMPUTER SCIENCE AND ENGINEERING, MINOR IN TRANSLATIONAL SENSORY SCIENCE

Sept. 2020 - May 2026 (estimated)

- Relevant Coursework: Motion and Planning in Games, 3D Computer Vision, Fundamentals of Machine Learning, Applied Motion Capture and Movement Analysis Technology, Advanced Computer Graphics, Computational Matrix Theory, Human Computer Interaction and User Interface Technology
- Associated Honors: National Science Foundation Graduate Training Program (NSF NRT) in Translational Sensory Science, Diversity of Viewpoints and Experiences Fellowship (DOVE)

### Aquincum Institute of Technology

Budapest, Hungary

STUDY ABROAD IN COMPUTER ENGINEERING

Jan. 2017 - May 2017

- Relevant coursework: Computer Vision Applications for Digital Cinema, Mobile Development

### Portland Community College

Portland, OR

SUPPLEMENTAL STUDY IN MULTIMEDIA DESIGN

Jun. 2016 - Dec. 2016

- Relevant coursework: Multimedia Design, Graphics for Multimedia, 3D Animation and Modeling
- Associated Honors: Dean's List

### Lewis & Clark College

Portland, OR

B.A. IN COMPUTER SCIENCE AND MATHEMATICS, MINOR IN PHYSICS

Aug. 2014 - May 2018

- Relevant coursework: Computer Graphics, Algorithms and Data Structures, Linear Algebra, Human-Computer Interaction
- Associated Honors: Robert B. Pamplin Honors Society of Fellows, Pi Mu Epsilon Oregon Eta Chapter President, Gender Minorities in STEM Club Founder and Leader (GEMSTEM), Dean's List

## Select Projects

### Unsupervised Deep-Learning for Motion Phenotyping

Minneapolis, MN

UNIVERSITY OF MINNESOTA - TWIN CITIES, APPLIED MOTION LAB

Jan 2024 - Present

- Unsupervised multidimensional clustering of raw and reduced motion data to identify key movement strategies with convolutional graph neural networks and contrastive learning for dimensionality reduction in **PyTorch**.
- Exploratory analysis to associate clusters with demographics and conditions for the identification of previously unknown subgroups.
- Methods validated on arm movement of typically developing people and full-body movement of at-risk infants in the NICU.

### Markerless Motion Capture in AR Game for Assessment of Bilateral Arm Use in Children

Minneapolis, MN

UNIVERSITY OF MINNESOTA - TWIN CITIES, APPLIED MOTION LAB AND NEUROREHABILITATION ACROSS THE LIFESPAN LAB

Jan 2023 - Present

- Video-based reaching game designed to induce unilateral and bilateral reaching in children without restriction built in **Unity** with pose extracted by **MediaPipe**.
- Motion capture data and game performance metrics collected from 151 typically developing children at the 2023 MN State Fair and from 400 people at the 2024 MN State Fair.
- Typical development performance markers and unique movement patterns identified across ages and genders with statistical significance.
- Associated papers published in Human Movement Science (Aug 2024) and under review for the Conference on Computer Animation and Social Agents (CASA).

### Video-Based Analysis Tool for Assessment of Bilateral Arm Use in Children with Hemiparetic Cerebral Palsy

Minneapolis, MN

UNIVERSITY OF MINNESOTA - TWIN CITIES, APPLIED MOTION LAB AND NEUROREHABILITATION ACROSS THE LIFESPAN LAB

Sept 2021 - Present

- Data processing and visualization toolkit to support clinicians in evaluating bilateral arm use.
- Analysis and associated GUI built in Python to handle pose estimations of the upper body from OpenPose.
- Deep machine learning assisted gesture annotation in Python using PyTorch for further clinician analysis assistance.
- Associated poster published at AACPDM 2023 Annual Meeting.
- Supported by Stanford's Restore Center pilot funding.

### Biased RRT\* Sampling with Predicted Distributions

Minneapolis, MN

UNIVERSITY OF MINNESOTA - TWIN CITIES, APPLIED MOTION LAB

July 2020 - May 2021

- Neural network in **Python** using **PyTorch**, to predict distributions for biased sampling-based path-planning algorithms.
- Training and testing data with **C++** implementation of a 2D biased Rapidly Exploring Random Tree Star (RRT\*) algorithm.

### Detailed 3D Reconstruction of Massively Occluded Faces

Minneapolis, MN

UNIVERSITY OF MINNESOTA - TWIN CITIES, ADVANCED COURSEWORK

Jan 2021 - May 2021

- Re-implemented modern 3D facial reconstruction pipeline in **Python** to evaluate success on faces > 50% occluded.
- Applied generative adversarial network (GAN) for in-painting on depth disparity bump map generation to improve the generation of realistic fine-grain details..
- Analyzed potential impact of this method on digital avatar construction and facial recognition.

### Physically-based Simulation and Realtime Graphics Exploration

Minneapolis, MN

UNIVERSITY OF MINNESOTA - TWIN CITIES, ADVANCED COURSEWORK

Sept. 2020 - May 2021

- Developed physics-based cloth and 3D elastic solid simulations from scratch in **C++** using **OpenGL**.
- Implemented RRT\* anticipatory path planning algorithm with online collision avoidance for procedural crowd simulation in **C++**.
- Built 3D ray tracer from base principles using projective geometric algebra in **C++** with custom libraries.

### Google Moss Map

Portland, OR

DOWNSTREAM

Jun 2019 - Jan 2020

- Native windows application that displays and synchronizes artistic content across 2 LED walls that flank projection on a physical map.
- Personally developed framework, transitional pieces, as well as 3 of the 7 simulation based content modes which included randomly generated roots that grew naturally across the screen and a stream of moving particles shaped like the city blocks.
- Application successfully installed in Google's new 315 Hudson office lobby in New York.

### Low-Cost Optical Motion Capture System

Portland, OR

LEWIS & CLARK COLLEGE

Aug. 2017 - May 2018

- Computer vision pipeline that takes in video footage of markers moving in space and computes the 3D location of the markers to output.
- Developed image processing and linear algebra tools and framework written in C++.
- Included hour long technical talk and publication in the library of theses by Lewis & Clark College.

### Individual Calibration of Redirected Walking Thresholds in Virtual Reality

Los Angeles, CA

UNIVERSITY OF SOUTHERN CALIFORNIA, INSTITUTE OF CREATIVE TECHNOLOGIES MIXED REALITY LAB

May 2017 - May 2018

- Virtual reality (VR) system to implement and test statistical algorithms that individually calibrate perception thresholds when changes to the locomotion mapping in VR occurs.
- Resultant poster published IEEE VR 2018 and research paper published in ICAT-EGVE 2018.

## Industry Experience

### Computer Vision Research Intern

Remote

KITWARE

May 2021 - August 2021

- Developed evaluation metric library in **Python** for active learning parameter tuning of motion tracking system.
- Established randomized baselines for evaluating effectiveness of various models.
- Integrated development within large scale project and open source libraries including KWIVR.
- Presented my progress effectively to the client every two weeks and to the company as a whole at the end of the program.

### Junior Interactive Software Developer

Portland, OR

DOWNSTREAM LLC

May 2018 - July 2020

- Built innovative interactive native windows applications or augmented reality mobile applications for more than 12 high-profile clients.
- Designed effective, flexible, and maintainable code structure in a **C++** environment that fulfilled client and designer requested features.
- Implemented improvements to company-wide frameworks based in **Cinder**, an open source graphics framework wrapper for **OpenGL**.
- Managed support tickets, onsite and remote installations, and both client and department collaborations.

### Visiting Research Assistant

Los Angeles, CA

UNIVERSITY OF SOUTHERN CALIFORNIA INSTITUTE OF CREATIVE TECHNOLOGIES MIXED REALITY LAB

May 2017 - Aug. 2017

- Developed novel research question after thorough analysis of previous work in redirected walking and human perception in virtual reality.
- Designed comprehensive research strategies and a full virtual reality user study in **Unity**.
- Worked independently as well as collaboratively to set and complete intermediate goals.

## Teaching Experience

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### Instructor on Record - Applied Machine Learning (CSCI 4521)

UNIVERSITY OF MINNESOTA - TWIN CITIES, DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Minneapolis, MN

Jan. 2025 - Present

- Define learning objectives and course policies for large advanced undergraduate elective course.
- Prepare and deliver 28 content-rich lectures to 135 students.
- Develop assignments and assessments to capture key data science and machine learning engineering skills.
- Manage task distribution and completion for 4 teaching assistants with varying workload requirements.
- Facilitate office hours and offline questions effectively and compassionately.

### Teaching Assistant - Program Design and Development (CSCI 3081w)

UNIVERSITY OF MINNESOTA - TWIN CITIES, DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Minneapolis, MN

Aug. 2023 - Dec. 2024

- Develop coursework and evaluation metrics to capture key curriculum features for **C++** and **Java** based course in software engineering.
- Manage two lab sections per term with direct instruction for approx. 60 students.
- Facilitate office hours and offline questions.
- Collaborate with 20 other undergraduate and graduate teaching assistants to provide support to over 300 undergraduate students.

### Co-Director / Instructor for Machine Learning Summer Camp

UNIVERSITY OF MINNESOTA - TWIN CITIES, DEPARTMENT OF MATHEMATICS

Minneapolis, MN

2022 - Present

- Develop week long data science and machine learning curriculum for over 50 high school students each summer.
- Manage and meet fundraising, enrollment, and curriculum goals.
- Oversee 3 other instructors, 4 teaching assistants, and 8 volunteers across two week-long camps.
- Successfully teach and mentor 15 students through their own independent machine learning projects in **Python** using **SciKit-Learn**.

### Peer Tutor and Teaching Assistant

LEWIS AND CLARK COLLEGE, DEPARTMENT OF MATHEMATICS AND COMPUTER SCIENCE

Portland, OR

Jan. 2015 - May 2018

- Tutored students in a self-directed atmosphere to strengthen concepts taught in math, computer science, and physics classes
- Facilitated office hours and graded labs for Java and Object-Oriented Programming based introductory course.
- Assisted professor in developing future coursework.

## Service

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2023 **Co-Chair**, Center for Applied Translational Sensory Science Spring Research Symposium,  
University of Minnesota

Minneapolis, MN

2021 **Founding Member**, Career Technical Education Advisory Board for Software Development and Digital Art,  
Da Vinci Charter Academy

Davis, CA

## Awards

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2018 **Honorable Mention, Collegiate Award**, National Center for Women and Information Technology (NCWIT)

2017 **Scholarship Recipient**, Grace Hopper Celebration

## Funding

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2023-2024 **Pilot Project Grant**, Stanford University Restore Center

2021-2023 **Research Trainee Graduate Training Program in Sensory Science**, National Science Foundation (NSF-NRT)

2020-2021 **Diversity of Views and Experiences Fellowship**, University of Minnesota Graduate School Diversity Office

## Publications

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Ziccardi, S., Chavis, Z., Hawe, R., & Guy, S.J. (*Under review*) **Reaching Motion Characterization Across Childhood via Augmented Reality Games**. In arXiv preprint [*submitted to Computer Animation and Social Agents (CASA), ACM*]. .

Ziccardi, S., Timanus, S., Ashrafzadehkian, G., Guy, S.J., & Hawe, R. (2024 August) **Characterization of bilateral reaching development using augmented reality games**. In Human Movement Science. Volume 96.

Hutton, C., Ziccardi, S., Medina, J., & Rosenberg, E. S. (2018 March) **Please don't puke: early detection of severe motion sickness in VR**. In 2018 IEEE Conference on Virtual Reality and 3D User Interfaces (VR). IEEE.

Hutton, C., Ziccardi, S., Medina, J., & Rosenberg, E. S. (2018 ) **Individualized Calibration of Rotation Gain Thresholds for Redirected Walking**. In ICAT-EGVE. ACM.

## Conference Presentations

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**Ziccardi, S.**, Raddatz, K., Guy, S., & Hawe, R. (2023 September) **Video-Based Analysis of Bilateral Arm Use in Children with Hemiparetic Cerebral Palsy (demonstration poster)**. In 2023 Annual Meeting. AACPDM.

**Ziccardi, S.**, Raddatz, K., Guy, S., & Hawe, R. (2022 April) **Vision-Based Analysis of Asymmetric Subject Gestures During Clinical Observation (poster)**. In Grad Cohort Workshop for Women. CRA-WP.